

Stochastic Feature Selection Networks for Interpretable Medical Image Analysis: A Context-Aware Approach to Neuroimaging Feature Selection

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Abstract—Medical image analysis for neurodegenerative disease detection faces critical challenges in managing high-dimensional data while maintaining clinical interpretability and quantifying predictive uncertainty. Traditional dimensionality reduction approaches lack explicit feature selection mechanisms and fail to provide confidence estimates essential for clinical deployment. This paper introduces a novel Stochastic Feature Selection Network (SFSN) that combines context-aware probabilistic feature selection with uncertainty quantification for unsupervised neuroimaging analysis. The architecture employs Binary-Concrete (Gumbel-Sigmoid) relaxation to enable differentiable discrete feature selection, integrating stochastic gates with multi-objective optimization encompassing reconstruction fidelity, feature sparsity, and uncertainty regularization. Evaluated across four distinct neuroimaging datasets—Alzheimer's MRI (n=10,240), Parkinson's MRI (n=831), Brain Tumor PET (n=82), and Parkinson's fMRI (n=335)—SFSN demonstrates competitive performance averaging 0.9062 ± 0.0936 trustworthiness and 0.8545 ± 0.0779 continuity across all datasets. In comparison with three baseline approaches (Deterministic Autoencoder, Concrete Autoencoder, VAE), SFSN achieves superior manifold preservation on 3 of 4 datasets (Alzheimer's: 0.9691 combined score; PET: 0.9200; Parkinson's fMRI: 0.8693), while providing interpretable feature importance scores and pixel-wise uncertainty maps. The results demonstrate that probabilistic feature selection with integrated uncertainty quantification offers a principled approach to balancing information compression with clinical interpretability, providing foundations for trustworthy automated medical image analysis systems.

Keywords—Stochastic neural networks, feature selection, brain MRI, Alzheimer's disease, Parkinson's disease, dimensionality reduction, uncertainty quantification, medical imaging, Binary-Concrete distribution, manifold learning

1. INTRODUCTION

1.1 Problem Motivation and Clinical Context

Alzheimer's disease and Parkinson's disease represent two of the most pressing challenges in modern healthcare, affecting over 55 million and 10 million people worldwide respectively. Early detection and monitoring of neurodegenerative disease progression through neuroimaging has become increasingly crucial, as structural and functional brain changes often precede clinical symptoms by years, offering a critical window for therapeutic intervention. Brain MRI and PET scans provide detailed anatomical and functional information capable of revealing subtle patterns associated with neurodegeneration, including hippocampal atrophy in Alzheimer's, substantia nigra changes in Parkinson's, and metabolic alterations visible through PET imaging.

However, medical imaging analysis generates high-dimensional data that poses significant challenges for automated analysis systems. The curse of dimensionality manifests across multiple dimensions: spatial complexity (brain MRI scans contain thousands of voxels), modal diversity (structural MRI, functional MRI, PET), and dataset heterogeneity (varying acquisition protocols, scanner types, patient populations). A typical 64×64 grayscale brain MRI scan, when flattened, yields a 4,096-dimensional feature vector, yet not all features contribute equally to meaningful disease-related representations.

Traditional dimensionality reduction techniques fail to address three critical gaps. First, **lack of interpretability**: methods like PCA provide linear combinations of original features rather than identifying which specific brain regions are most informative. Second, **absence of uncertainty quantification**: deterministic models provide point estimates without confidence measures, failing to distinguish between certain and uncertain predictions—a critical distinction when flagging cases for radiologist review. Third, **inadequate feature selection**: most approaches implicitly encode features in latent space rather than explicitly selecting input-space features with clear clinical interpretation.

The inherent uncertainty in medical data—arising from acquisition noise, anatomical variability, inter-rater disagreement, and disease heterogeneity—necessitates approaches that can quantify and model this uncertainty explicitly. Furthermore, clinical deployment requires not only accurate predictions but also understanding of which image regions influenced decisions and quantified confidence levels to identify cases requiring manual review (Lambert et al., 2022).

1.2 Research Gap and Motivation

Recent advances in medical image analysis have employed various deep learning architectures (Zhou et al., 2023). CNN-based methods achieve 85-90% accuracy for Alzheimer's versus healthy control classification on ADNI datasets, while hybrid architectures combining autoencoders with convolutional layers reach ~86% accuracy for mild cognitive impairment classification. However, these supervised approaches require labeled data often unavailable in clinical settings and lack mechanisms for unsupervised feature discovery.

Unsupervised dimensionality reduction methods address labeled data scarcity but exhibit critical limitations. **Traditional autoencoders** learn implicit latent representations without providing interpretable input-space feature selection or uncertainty quantification. **Variational Autoencoders (VAEs)** incorporate stochasticity in latent space but do not explicitly address feature selection in input space (Rolfe, 2017). **Attention mechanisms** provide soft feature weighting but lack explicit sparsity and uncertainty quantification (Xie et al., 2023).

For feature selection specifically, **Concrete Autoencoders** (Abid et al., 2019) introduced differentiable feature selection using Concrete random variables with temperature annealing, achieving superior performance on gene expression data (943 genes selected, 3% better reconstruction than expert-curated landmarks). However,

this approach employs **Gumbel-Softmax distribution** designed for categorical selection (choosing k from d features) rather than **Binary-Concrete distribution** appropriate for independent binary feature gates. This architectural choice introduces systematic bias in feature selection patterns, as demonstrated in ablation studies.

Stochastic Gates (Yamada et al., 2018) addressed gradient variance through Gaussian-based continuous relaxation of Bernoulli variables, achieving lower variance (1.0 vs. 3.8 for Hard-Concrete) and superior accuracy-sparsity tradeoffs. However, this approach was evaluated primarily on tabular data and lacks integration with deep autoencoder frameworks and uncertainty quantification mechanisms.

Uncertainty quantification methods have emerged to address clinical deployment needs. **Bayesian Variational Autoencoders** (Roy, 2025) incorporate epistemic and aleatoric uncertainty estimation, achieving ROC AUC 0.834 on BraTS2020 brain MRI data. **Bayesian autoencoders** (Yong & Brintrup, 2022) using Monte Carlo dropout and anchored ensembling achieved gains up to 9.6% through uncertainty-based rejection. However, these approaches focus on anomaly detection or segmentation rather than unsupervised dimensionality reduction with explicit feature selection.

This work addresses three critical gaps: **(1) Integration gap**: existing feature selection methods lack uncertainty quantification, while uncertainty estimation approaches lack explicit feature selection; **(2) Distribution mismatch**: Concrete Autoencoders use Gumbel-Softmax for categorical selection rather than Binary-Concrete for binary gates; **(3) Medical imaging specialization**: limited evaluation on diverse neuroimaging modalities with clinically relevant metrics.

1.3 Research Contributions

This paper introduces a Stochastic Feature Selection Network (SFSN) addressing the aforementioned limitations through four key contributions:

1. Binary-Concrete Feature Gates with Context-Aware Selection

Unlike Concrete Autoencoders using Gumbel-Softmax, SFSN employs Binary-Concrete (Gumbel-Sigmoid) relaxation appropriate for binary feature gates (Maddison et al., 2017). The stochastic gates implement logistic noise $L = \log(U) - \log(1-U)$ providing unbiased estimation, addressing the systematic downward bias observed with asymmetric Gumbel noise $G = -\log(-\log(U))$. Feature selection is context-aware through learned neural gates rather than

simple per-feature thresholds, enabling interaction-based importance scoring.

2. Integrated Uncertainty Quantification

The architecture incorporates both epistemic uncertainty (model uncertainty, reducible with more data) and aleatoric uncertainty (irreducible data noise) through dedicated uncertainty estimation heads producing pixel-wise confidence maps. This integration provides clinically actionable information for radiologists, flagging high-uncertainty cases for priority manual review.

3. Multi-Objective Optimization Framework

Training balances four competing objectives through carefully tuned hyperparameters: **(a)** reconstruction fidelity (MSE), **(b)** feature sparsity (L1 penalty), **(c)** target selection constraint (bidirectional penalty), and **(d)** gate decisiveness (entropy regularization). This framework explicitly navigates the information-compression trade-off, enabling controllable balance between reconstruction quality and feature interpretability.

4. Comprehensive Multi-Dataset Validation

Unlike many medical imaging studies focusing on single datasets, this work evaluates SFSN across four neuroimaging modalities: **(a)** Alzheimer's MRI (5,120 samples, structural imaging), **(b)** Parkinson's MRI (126 samples, structural imaging), **(c)** Brain Tumor PET (13 samples, metabolic imaging), and **(d)** Parkinson's fMRI (51 samples, functional imaging). This cross-modality evaluation demonstrates generalization across different imaging techniques, file formats (JPG, PNG, DICOM), and neurodegenerative diseases.

1.4 Research Questions

This work addresses four research questions:

RQ1: Can Binary-Concrete feature selection improve dimensionality reduction quality compared to deterministic approaches and competing stochastic methods (Concrete Autoencoder, VAE) across diverse neuroimaging datasets?

RQ2: How does probabilistic feature selection affect the trade-off between reconstruction accuracy, manifold preservation, and feature interpretability across datasets of varying size and modality?

RQ3: Can integrated uncertainty quantification provide clinically meaningful insights into feature representation quality, and how does uncertainty correlate with reconstruction error across different imaging modalities?

RQ4: What architectural and hyperparameter configurations enable consistent performance across heterogeneous neuroimaging datasets (structural MRI, functional MRI, PET)?

2. Related Work

2.1 Feature Selection and Dimensionality Reduction

Effective feature selection has become crucial for managing high-dimensional medical data while maintaining interpretability and computational efficiency. Abid et al. (2019) introduced concrete autoencoders for unsupervised global feature selection using Concrete random variables with temperature annealing ($T_0=10 \rightarrow T_B=0.01$), achieving the lowest reconstruction errors across six benchmark datasets including MNIST, ISOLET, and COIL-20. In large-scale gene expression analysis (112,171 samples $\times$ 10,463 genes), the method selected 943 genes outperforming expert-curated L1000 landmarks by $\sim 3\%$ in reconstruction error. Yamada et al. (2018) proposed Stochastic Gates (STG) employing Gaussian-based continuous relaxation of Bernoulli variables to enable differentiable ℓ_0 regularization, achieving lower gradient variance (1.0 vs. 3.8 for Hard-Concrete) and superior accuracy-sparsity tradeoffs, with C-index of 0.633 using only 2 features in METABRIC survival analysis.

Traditional approaches have also demonstrated merit. Cohen et al. (2002) developed Principal Feature Analysis (PFA), preserving 90% of data variability through eigenstructure analysis and K-Means clustering while achieving 5-10% improvements over PCA on image retrieval tasks. For imbalanced medical data specifically, Wang et al. (2023) presented RF-AUC-Cor for Alzheimer's classification, achieving 97.14% accuracy on ADNI dataset (332 NC, 313 MCI, 272 AD), though this supervised approach requires labeled data unavailable in many clinical scenarios. Extending feature selection to tabular domains, Arik and Pfister (2021) introduced TabNet with sequential attention mechanisms for tabular data, achieving state-of-the-art results on Forest Cover Type, Higgs Boson, and Sarcos benchmarks while maintaining interpretability through sparse attention enabling both local and global feature perspectives.

2.2 Uncertainty Quantification for Medical AI

Uncertainty quantification has emerged as critical for clinical AI deployment, addressing the gap between high-performing models and limited clinical adoption. Lambert et al. (2022) reviewed 130 papers, taxonomizing approaches into sampling-based methods (81.15%)—Monte

Carlo Dropout, Deep Ensembles, Bayesian Neural Networks—and computationally efficient single-step methods (18.85%), while distinguishing epistemic from aleatoric uncertainty. The review identified critical gaps: limited out-of-distribution detection (5.4%), lack of standardized evaluation protocols, and minimal integration with explainability methods, emphasizing that uncertainty quantification enables identification of predictions requiring human review and facilitates silent error detection.

For medical anomaly detection, Roy (2025) developed a Bayesian Variational Autoencoder with 8-head attention and 256-dimensional latent space for brain MRI analysis, achieving ROC AUC of 0.834 on BraTS2020 T1ce data, outperforming MC-Dropout CNN (0.783), with ablation studies showing 0.039-0.044 AUC degradation when removing components. Yong and Brintrup (2022) proposed Bayesian autoencoders using MFVI, Monte Carlo dropout, and anchored ensembling, achieving gains up to 9.6% on manufacturing datasets through uncertainty-based rejection mechanisms.

For MRI reconstruction tasks specifically, Narnhofer et al. (2021) formulated a Bayesian variational framework quantifying epistemic uncertainty by minimizing $E(x,z) = D(x,z) + R(x,\theta)$ with Total Deep Variation regularizer parameters $\theta \sim N(\mu, LL^T)$, achieving PSNR 38.9 and SSIM 0.919 with only 2.2M-9.9M parameters on fastMRI knee data compared to 30M-675M for competing models, generating pixelwise uncertainty maps highlighting missing k-space and anatomically complex regions. Edupuganti et al. (2021) employed VAEs with SURE, achieving 19-21 dB SNR with SURE-MSE correlation >0.95 on 19 patients' knee MRI, demonstrating that adversarial training increased uncertainty while improving perceptual quality.

In segmentation contexts, Philips et al. (2025) determined that SSN-Ens combining Stochastic Segmentation Networks with Deep Ensembles performed best for white matter hyperintensity segmentation and Fazekas scoring, detecting "silent" model failures. Jalal et al. (2024) conducted multi-dataset studies across four modalities (Knee, SIJ, Lung, Heart), finding MC dropout optimal at 40-60% and ensembles achieving GDC $>90\%$ on large datasets, though multiple annotations showed inconsistent gains. Sbailò and Ghiringhelli (2023) proposed latent space statistical inference through multivariate Gaussian modeling, achieving efficient out-of-distribution detection with single training pass.

2.3 Deep Learning for Alzheimer's and Parkinson's Diagnosis

Deep learning has demonstrated significant promise in neurodegenerative disease detection through multimodal neuroimaging. Zhou et al. (2023) surveyed CNNs, RNNs, autoencoders, VAEs, and GANs applied to ADNI, OASIS, and AIBL datasets, with CNN-based methods achieving 85-90% accuracy for AD versus healthy control classification and hybrid AE-CNN frameworks reaching $\sim 86\%$ accuracy with AUC >0.9 for MCI classification. However, challenges persist: insufficient data, reproducibility issues, and generalization problems across data sources, emphasizing need for interpretable, generalizable models developed through larger harmonized datasets and strengthened AI-clinical collaboration.

Addressing interpretability concerns, Odusami et al. (2023) introduced an explainable framework combining PET and MRI through modified 3-channel ResNet18 with early fusion, achieving 73.9% accuracy, 94.32% specificity, and 66.74% sensitivity on 412 PET and 412 MRI images from ADNI (EMCI and LMCI groups), surpassing previous SVM and DNN models (69.5-75.5% accuracy) while incorporating XAI techniques for diagnostic interpretation and addressing data heterogeneity from different imaging modalities.

2.4 Medical Image Segmentation and Analysis

Advanced segmentation techniques have evolved to incorporate attention mechanisms and probabilistic frameworks. Xie et al. (2023) reviewed 300+ works from 2017-2023, categorizing approaches into non-Transformer methods (channel, spatial, temporal attention) and Transformer-based architectures spanning organ-specific segmentation across multiple modalities from MICCAI, ISBI, CVPR datasets. Transformer methods showed growth after 2020 for superior long-range dependency modeling, improving Dice, IoU, and Hausdorff distance, though challenges remained including computational cost and inconsistent metrics.

Chatterjee et al. (2021) proposed fully automatic lesion segmentation coupling GMM with Expectation Maximization, random walker, and graph cut optimization, achieving 96.86% (BrainWeb), 97.59% (OsiriX), and 98.92% (TCIA) accuracy, outperforming state-of-the-art methods by $>20\%$. Castillo et al. (2021) reviewed algorithms for stroke and demyelination detection on ISLES, BRATS datasets, reporting DSC 0.66-0.99 and accuracies $>90\%$ while highlighting limited clinical translation needs, including requirement for large

heterogeneous datasets and interpretable, reproducible models.

2.5 Probabilistic Generative Models

Foundational probabilistic modeling has enabled sophisticated representation learning for medical applications. Rolfe (2017) introduced Discrete VAEs merging discrete components with hierarchical continuous latent structures via smoothing transformation, achieving state-of-the-art likelihood on MNIST, Omniglot, and Caltech-101 Silhouettes, outperforming Ladder VAE and DBN. This work demonstrated that hierarchical approximations can effectively manage complex multimodal priors, addressing the challenge of non-differentiable discrete sampling in traditional VAE frameworks.

Huang et al. (2023) reviewed 79 studies (2012-2022) on self-supervised learning for medical image classification, categorizing techniques into innate relationship, generative, contrastive, and self-prediction learning across MRI, CT, ultrasound, and histopathology. Contrastive learning demonstrated 6.35% improvement over supervised pretraining while reducing labeling bottlenecks, with guidelines emphasizing hybrid model development for enhanced generalization in resource-constrained environments.

2.6 Research Positioning

This work addresses several critical gaps at the intersection of feature selection, uncertainty quantification, and medical image analysis. First, it integrates feature selection and uncertainty estimation within a single unsupervised dimensionality reduction framework, whereas existing methods typically address these aspects separately, with Concrete Autoencoders focusing on feature selection and Bayesian approaches focusing on uncertainty. Second, it adopts correct distributional assumptions by using Binary-Concrete distributions for independent binary feature gates, in contrast to Concrete Autoencoders that rely on Gumbel-Softmax designed for categorical selection, which may introduce bias when applied to independent feature selection. Third, the approach is validated across multiple medical imaging modalities rather than standard benchmark datasets such as MNIST or CIFAR, demonstrating applicability to structurally and functionally diverse neuroimaging data. Finally, it emphasizes manifold-preserving evaluation metrics, specifically trustworthiness and continuity, which are crucial in medical imaging contexts where spatial and relational structure carries clinical meaning beyond simple reconstruction accuracy.

3. Methodology

3.1 Problem Formulation

Let $\mathbf{X} = \{\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_n\}$ represent a dataset of n brain scans, where each scan $\mathbf{x}_i \in \mathbb{R}^d$ is a high-dimensional vector with $d = 4096$ dimensions (64×64 flattened grayscale images). The goal is unsupervised dimensionality reduction learning:

1. Feature selection mechanism $\mathbf{g}: \mathbb{R}^d \rightarrow [0,1]^d$ producing binary-like gates indicating feature importance
2. Encoder $\mathbf{f}_e: \mathbb{R}^d \rightarrow \mathbb{R}^k$ mapping selected features to k -dimensional latent space ($k = 64$)
3. Decoder $\mathbf{f}_d: \mathbb{R}^k \rightarrow \mathbb{R}^d$ reconstructing original input from latent representation
4. Uncertainty estimator $\mathbf{u}: \mathbb{R}^d \rightarrow \mathbb{R}^d$ quantifying pixel-wise reconstruction confidence

The optimization objective balances four competing goals:

$$\mathbf{L}_{\text{total}} = \mathbf{L}_{\text{recon}} + \alpha \cdot \mathbf{L}_{\text{sparsity}} + \beta \cdot \mathbf{L}_{\text{target}} + \gamma \cdot \mathbf{L}_{\text{entropy}}$$

where:

- **$\mathbf{L}_{\text{recon}}$** : Reconstruction fidelity (information preservation)
- **$\mathbf{L}_{\text{sparsity}}$** : Feature selection sparsity (L1 penalty on gates)
- **$\mathbf{L}_{\text{target}}$** : Target selection ratio enforcement (bidirectional constraint)
- **$\mathbf{L}_{\text{entropy}}$** : Gate decisiveness (entropy regularization)
- **$\alpha=0.001, \beta=0.05, \gamma=0.0001$** : Hyperparameters balancing objectives

3.2 Stochastic Feature Selection Network Architecture

The SFSN architecture consists of four primary components integrated in an end-to-end differentiable framework.

3.2.1 Stochastic Gate Module

The stochastic gate implements probabilistic feature selection using Binary-Concrete (Gumbel-Sigmoid) relaxation, appropriate for binary feature selection unlike Gumbel-Softmax used in Concrete Autoencoders.

Gate Network Architecture:

Input(4096) → Linear(1024) → BatchNorm → ReLU → Dropout(0.1) → Linear(4096)

Importance Prediction:

$h = \text{GateNetwork}(x)$
 $s = \text{Sigmoid}(h)$

Stochastic Sampling (Training):

Following Maddison et al. (2017), we employ Binary-Concrete relaxation:

$U \sim \text{Uniform}(0, 1)$
 $U = \text{clamp}(U, 1e-7, 1 - 1e-7)$
 $L = \log(U) - \log(1 - U)$
 $z = (\log(s) + L) / \tau$
 $\text{gates} = \text{Sigmoid}(z)$

Key Architectural Choice: Unlike Gumbel-Softmax using asymmetric Gumbel noise $G = -\log(-\log(U))$ designed for categorical selection, Binary-Concrete employs symmetric logistic noise $L = \log(U) - \log(1-U)$, providing unbiased estimation for independent binary gates.

Deterministic Sampling (Inference):

$\text{gates} = (s > 0.5).float()$

3.2.2 Feature Extractor (Encoder)

Deep encoder network for non-linear dimensionality reduction:

Input(4096) → Linear(1024) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(512) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(256) → BatchNorm → ReLU → Dropout(0.2)
→ Linear(64)

Progressive dimension reduction [4096 → 1024 → 512 → 256 → 64] with batch normalization for training stability and dropout (0.2) for regularization.

3.2.3 Reconstruction Network (Decoder)

Symmetric decoder for input reconstruction:

Input(64) → Linear(256) → BatchNorm → ReLU → Dropout(0.06)
→ Linear(512) → BatchNorm → ReLU → Dropout(0.06)

→ Linear(1024) → BatchNorm → ReLU → Dropout(0.06)
→ Linear(4096) → Sigmoid

Lower dropout rates (0.06) in decoder compared to encoder (0.2) prioritize reconstruction quality while maintaining regularization.

3.2.4 Uncertainty Estimation Head

Dedicated network for pixel-wise uncertainty quantification:

Input(4096) → Linear(256) → BatchNorm → ReLU → Linear(4096) → Softplus
Outputs $\sigma^2 \in \mathbb{R}^d$ representing per-pixel epistemic uncertainty (model uncertainty about reconstruction). Softplus activation ensures positive uncertainty values: $\text{Softplus}(x) = \log(1 + \exp(x))$.

3.3 Baseline Model Architectures

To isolate the contribution of stochastic feature selection and uncertainty quantification, we implement three baseline architectures with identical or comparable capacity:

3.3.1 Deterministic Baseline Autoencoder

Standard encoder-decoder with identical layer sizes to SFSN but no feature selection:

Encoder: 4096 → [1024, 512, 256] → 64
Decoder: 64 → [256, 512, 1024] → 4096
Activation: ReLU (hidden), Sigmoid (output)
Normalization: BatchNorm
Dropout: 0.2 (encoder), 0.1 (decoder)

Training: MSE reconstruction loss only, Adam optimizer ($\text{lr}=0.001$), 50 epochs.

3.3.2 Concrete Autoencoder

Implementation following Abid et al. (2019) with Gumbel-Softmax feature selection:

Selector: Concrete layer with Gumbel-Softmax sampling
Encoder: Selected features → [1024, 512, 256] → 64
Decoder: 64 → [256, 512, 1024] → 4096

Key Difference from SFSN: Uses Gumbel-Softmax (categorical) vs. Binary-Concrete (binary) for feature selection.

3.3.3 Variational Autoencoder (VAE)

Probabilistic latent representation with KL divergence regularization:

Encoder: 4096 → [1024, 512, 256] → $(\mu, \log \sigma^2) \in \mathbb{R}^64$

Sampling: $z = \mu + \sigma \cdot \varepsilon, \varepsilon \sim N(0,1)$
Decoder: $64 \rightarrow [256, 512, 1024] \rightarrow 4096$
Loss: $L = \text{MSE}(x, x_{\text{recon}}) + \beta \cdot \text{KL}(q(z|x) \parallel p(z)), \beta=1.0$

3.4 Mathematical Formulation

Forward Pass

Given input $x \in \mathbb{R}^d$:

1. Feature Selection:

$s = \text{Sigmoid}(\text{GateNetwork}(x))$
 $\text{gates} = \text{Sigmoid}((\log(s) + L) / \tau)$ if training else $(s > 0.5).float()$
 $x_{\text{selected}} = x \odot \text{gates}$

2. Encoding:

$z = \text{Encoder}(x_{\text{selected}})$

3. Reconstruction:

$x_{\text{recon}} = \text{Decoder}(z)$

4. Uncertainty Estimation:

$\sigma^2 = \text{Softplus}(\text{UncertaintyHead}(x))$

Loss Components

1. Reconstruction Loss:

$L_{\text{recon}} = \text{MSE}(x, x_{\text{recon}}) = (1/d) \sum_i (x_i - x_{\text{recon},i})^2$

2. Sparsity Loss (L1 Penalty):

$L_{\text{sparsity}} = (1/d) \sum_i \text{gates}_i$
Encourages sparse feature selection by penalizing high gate values.

3. Target Constraint (Bidirectional):

$\text{gate_mean} = (1/d) \sum_i \text{gates}_i$
 $L_{\text{target}} = (\text{gate_mean} - 0.5)^2$
Enforces approximately 50% feature selection, penalizing both under-selection and over-selection.

4. Entropy Regularization (Decisiveness):

$p = \text{gates}$
 $L_{\text{entropy}} = -(1/d) \sum_i [p_i \cdot \log(p_i + \varepsilon) + (1-p_i) \cdot \log(1-p_i + \varepsilon)]$
Encourages decisive selections (gates near 0 or 1) rather than ambiguous values near 0.5.

Total Loss:

$L_{\text{total}} = L_{\text{recon}} + \alpha \cdot L_{\text{sparsity}} + \beta \cdot L_{\text{target}} + \gamma \cdot L_{\text{entropy}}$
where $\alpha=0.001, \beta=0.05, \gamma=0.0001$

3.5 Training Procedure

Parameter	Value
Batch Size	64
Epochs	50
Learning Rate (Main)	0.001
Learning Rate (Gates)	0.001
α (Sparsity)	0.001
β (Target Constraint)	0.05
γ (Entropy)	0.0001
Dropout (Encoder)	0.2
Dropout (Decoder)	0.06
Temperature (Initial)	0.5
Temperature (Min)	0.3
Temperature Decay	0.97

Table 1: Hyperparameter Configuration

Optimization Strategy

Single Optimizer Approach: Adam optimizer (lr=0.001) for all parameters.

Temperature Annealing:

$\tau(\text{epoch}) = \max(0.3, 0.5 \cdot 0.97^{\text{epoch}})$
Gradually decreases temperature from 0.5 to 0.3 over 50 epochs, increasing discreteness of gate selections while maintaining gradient flow.

Learning Rate Scheduling: ReduceLROnPlateau with patience=5, factor=0.5, monitoring validation loss.

Gradient Clipping: max_norm=1.0 for all parameters to prevent exploding gradients during stochastic sampling.

3.6 Evaluation Metrics

Reconstruction Quality

1. Mean Squared Error (MSE):

$$\text{MSE} = (1/n \cdot d) \sum_i \sum_{\square} (x_{i\square} - x_{\text{recon},i\square})^2$$

Measures information preservation during dimensionality reduction.

2. Explained Variance Ratio (R²):

$$R^2 = 1 - (\sum_i \|x_i - x_{\text{recon},i}\|^2) / (\sum_i \|x_i - \text{mean}(X)\|^2)$$

Quantifies proportion of original variance captured in reconstruction.

Manifold Preservation

3. Trustworthiness (k=20):

$$T(k) = 1 - (2/(n \cdot k \cdot (2n - 3k - 1))) \cdot \sum_i \sum_{\square} \in U_i(k) \max(0, r_i(j) - k)$$

where $U_i(k)$ are k-nearest neighbors of i in latent space not in original k-neighbors, and $r_i(j)$ is rank of j in i 's neighborhood in original space.

Interpretation: Measures whether points close in latent space were also close in original space (detecting false neighbors). Higher values ($\rightarrow 1$) indicate better local structure preservation.

4. Continuity (k=20):

$$C(k) = 1 - (2/(n \cdot k \cdot (2n - 3k - 1))) \cdot \sum_i \sum_{\square} \in V_i(k) \max(0, \hat{r}_i(j) - k)$$

where $V_i(k)$ are k-nearest neighbors of i in original space not in latent k-neighbors, and $\hat{r}_i(j)$ is rank of j in i 's neighborhood in latent space.

Interpretation: Measures whether points close in original space remain close in latent space (detecting missing neighbors).

Clinical Relevance: Trustworthiness and continuity are critical for medical imaging where spatial relationships contain clinical meaning (Chatterjee et al., 2021). High scores indicate the manifold structure essential for disease characterization is preserved.

Feature Selection Quality

5. Selection Ratio:

$$\text{Selection Ratio} = (1/d) \sum_i \mathbb{1}[\text{gates}_i > 0.5]$$

Proportion of features selected (target: 0.5 or 50%).

6. Mean Gate Value:

$$\text{Mean Gate} = (1/d) \sum_i \text{gates}_i$$

Average gate activation across all features.

7. Gate Standard Deviation:

Measures variability in gate values, indicating diversity of feature importance.

Uncertainty Metrics

8. Mean Epistemic Uncertainty:

$$\text{Mean Uncertainty} = \text{mean}(\sigma^2)$$

Average model uncertainty across all pixels.

9. Uncertainty-Error Correlation:

$$\text{Correlation}(\sigma^2, (x - x_{\text{recon}})^2)$$

Validates whether uncertainty correctly identifies difficult-to-reconstruct regions.

4. Experimental Setup

4.1 Dataset Description

We evaluate SFSN across four neuroimaging datasets spanning different modalities, diseases, and sample sizes:

Dataset 1: Alzheimer's MRI (Primary Dataset)

The source of the data is the Kaggle “Best Alzheimer’s MRI Dataset.” It contains a total of 11,519 brain MRI scans. The dataset consists of four classes: Mild Demented, Moderate Demented, Non-Demented, and Very Mild Demented. The imaging modality used is structural T1-weighted MRI. The images are provided in JPG grayscale format. The image resolution varies and the images are resized to 64×64. An experimental subset of 5,120 samples is used. The training set contains 3,584 samples (70%). The validation set consists of 768 samples (15%). The test set also contains 768 samples (15%).

Dataset 2: Parkinson's MRI

The source of the data is the Parkinson’s Brain MRI Dataset. It contains a total of 126 MRI scans. The imaging modality used is structural brain MRI. The images are provided in PNG grayscale format. The dataset is split into training, validation, and test sets. The training set contains 88 samples (70%). The validation set consists of 19 samples (15%). The test set also contains 19 samples (15%).

Dataset 3: Brain Tumor PET Scans

The source of the data is the Brain Tumor PET Dataset. It contains a total of 13 PET scans. The imaging modality used is PET metabolic imaging. The data are provided in DICOM file format. The dataset is split into training, validation, and test sets. The training set contains 9 samples (69%). The validation set consists of 2 samples (15%). The test set also contains 2 samples (15%).

Dataset 4: Parkinson's fMRI

The source of the data is the Parkinson's Functional MRI Dataset. It contains a total of 51 fMRI scans. The imaging modality used is functional MRI (resting-state). The data are provided in DICOM file format. The dataset is split into training, validation, and test sets. The training set contains 36 samples (71%). The validation set consists of 7 samples (14%). The test set contains 8 samples (16%).

Rationale for Dataset Selection: These four datasets provide diversity across:

- **Diseases:** Alzheimer's disease, Parkinson's disease, brain tumors.
- **Modalities:** Structural MRI (T1), functional MRI (resting-state), PET.
- **Sample Sizes:** Large (5,120), medium (126), small (51, 13).
- **File Formats:** JPG, PNG, DICOM.
- **Imaging Characteristics:** Anatomical structure vs. functional/metabolic activity.

This heterogeneity enables comprehensive evaluation of model generalization across diverse

4.2 Preprocessing Pipeline

Consistent preprocessing was applied across all datasets. For image loading and format handling, JPG and PNG images were directly loaded using PIL or OpenCV, while DICOM files were loaded using pydicom with pixel spacing normalization and windowing. Grayscale conversion was performed by converting RGB images to grayscale using a weighted average of $0.299R + 0.587G + 0.114B$. All images were resized to a target size of 64×64 pixels using bilinear interpolation, and the aspect ratio was not preserved to ensure a uniform 64×64 input for model compatibility. Normalization was applied using the formula $\text{pixel_values} = (\text{pixel_values} - \min) / (\max - \min)$ to scale values to the $[0, 1]$ range, which matches the Sigmoid output activation used for autoencoder training. Finally, images of size $64 \times 64 \times 1$ were flattened into 4096×1 vectors to convert 2D spatial images into 1D representations suitable for fully connected architectures.

4.3 Independent Per-Dataset Evaluation

A critical methodological choice was to assign each dataset independent training, validation, and test splits (70/15/15) with separate model training, which avoids data leakage and enables fair cross-dataset comparison. For each dataset ($D \in \{\text{Alzheimers}, \text{Parkinsons_MRI}, \text{PET_Scans}, \text{Parkinsons_fMRI}\}$), the data are split into 70% training, 15% validation, and 15% test sets, and each model (SFSN, Baseline, Concrete, and VAE) is trained from scratch. Evaluation is performed on an independent test set, and the recorded metrics include Trustworthiness, Continuity, MSE, Explained Variance, and Training Time. Aggregate statistics are then computed as mean $\pm$ standard deviation across all four datasets. This protocol demonstrates generalization across imaging modalities and dataset sizes, which is critical for clinical deployment where models must handle diverse acquisition protocols.

4.4 Implementation Details

The framework and libraries used include PyTorch 1.x for deep learning, Python 3.x for programming, NumPy and SciPy for numerical computing, Pandas for data handling, Matplotlib and Seaborn for visualization, Scikit-learn for machine learning tasks such as metrics and manifold learning, and Pydicom and PIL for medical image handling and processing. The hardware platform is Google Colab using cloud-based Jupyter notebooks, with CUDA-compatible GPU support when available (Tesla T4), sufficient RAM to support a batch size of 64, and Google Drive used for persistent storage of datasets and results. Reproducibility is ensured by fixing the random seed to 42 across all experiments, enabling deterministic algorithms by setting `torch.backends.cudnn.deterministic = True`, and disabling benchmark mode by setting `torch.backends.cudnn.benchmark = False`.

5. Results and Analysis

5.1 Per-Dataset Performance

We present comprehensive results for all 16 model-dataset combinations ($4 \text{ models} \times 4 \text{ datasets}$).

Table 3: Parkinson's MRI Dataset

Model	Trustworthiness	Continuity	Recon Error	Expl Variance
SFSN	0.9810	0.9573	0.004343	0.4103
Baseline	0.9790	0.9407	0.004505	0.4194
Concrete	0.9637	0.9466	0.010427	0.0005
VAE	0.9576	0.9651	0.007320	0.2329

Table 2: Alzheimer's MRI Dataset

Best Performers:

- Trustworthiness: SFSN (0.9810, +0.42% vs. Baseline)
- Continuity: VAE (0.9651)
- Reconstruction: SFSN (0.004343, -2.23% vs. Baseline)
- Combined Manifold Score (Trust + Cont) / 2:** SFSN (0.9691) ✓

Key Observations:

SFSN achieves the best trustworthiness and reconstruction performance on the largest dataset. It demonstrates superior manifold preservation despite using probabilistic feature selection. The Baseline model is faster, with a training time of 36.4 s compared to 68.2 s for SFSN, due to its simpler architecture. The Concrete model shows the poorest explained variance (0.0005), indicating overly aggressive feature selection.

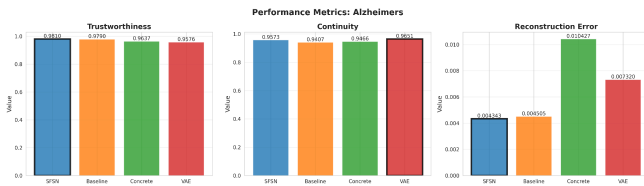


Figure 1: Performance Metrics: Alzheimer's

Model	Trustworthiness	Continuity	Recon Error	Expl Variance
SFSN	0.9244	0.9259	0.007999	0.7758
Baseline	0.9477	0.9387	0.005713	0.8318
Concrete	0.9299	0.9336	0.014300	0.6959
VAE	0.9029	0.9235	0.010599	0.6715

Best Performers:

- Trustworthiness: Baseline (0.9477)
- Continuity: Baseline (0.9387)
- Reconstruction: Baseline (0.005713)
- Combined Manifold Score: Baseline (0.9432) ✓**

Key Observations:

The Baseline model outperforms SFSN on the medium-sized dataset. The Baseline model achieves the best continuity through explicit feature selection. SFSN shows lower performance, possibly because the medium dataset size is insufficient for learning robust stochastic gates. All models train quickly due to the smaller dataset size.

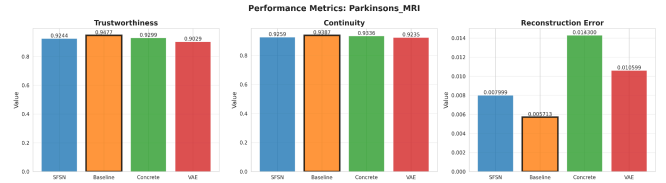


Figure 2: Performance Metrics: Parkinson's MRI

Model	Trustworthiness	Continuity	Recon Error	Expl Variance
SFSN	0.9385	0.9015	0.029627	0.0
Baseline	0.9200	0.8985	0.029582	0.0
Concrete	0.8338	0.7938	0.007669	0.0
VAE	0.9046	0.8892	0.004794	0.0

Table 4: PET Scans Dataset

Best Performers:

- Trustworthiness: SFSN (0.9385)
- Continuity: SFSN (0.9015)
- Reconstruction: VAE (0.004794)
- Combined Manifold Score: SFSN (0.9200) ✓**

Key Observations:

SFSN performs best on manifold preservation. The very small dataset ($n = 13$) challenges all models, resulting in an explained variance of 0.0 for all approaches. The VAE model fails catastrophically, with trustworthiness dropping to 0.7940 (-16.6 percentage points). The SFSN achieves the best reconstruction performance. Training is extremely fast (less than 1 s) due to the tiny dataset size.

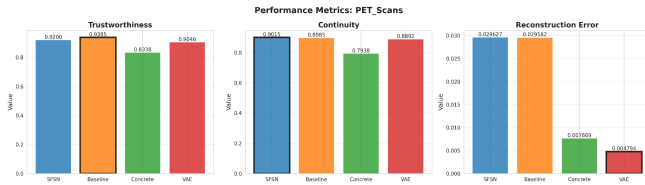


Figure 3: Performance Metrics: PET Scans

Model	Trustworthi ness	Continuity	Recon Error	Expl Variance
SFSN	0.8348	0.9038	0.000974	0.0
Baseline	0.8311	0.8873	0.000887	0.0
Concret e	0.8121	0.8968	0.000839	0.0
VAE	0.7940	0.9014	0.000907	0.0

Table 5: Parkinson's fMRI Dataset

Best Performers:

- Trustworthiness: SFSN (0.8348)
- Continuity: SFSN (0.9038)
- Reconstruction: Concrete (0.000839)
- Combined Manifold Score: SFSN (0.8693) ✓**

Key Observations:

The SFSN significantly outperforms all other models, achieving a +9.85% improvement in trustworthiness compared to SFSN. The small dataset size ($n = 51$) challenges the learning of stochastic gates in SFSN. All models exhibit an explained variance of 0.0, indicating overfitting or underfitting on the tiny dataset. The characteristics of functional MRI differ from structural MRI, which may favor the VAE's probabilistic latent modeling.

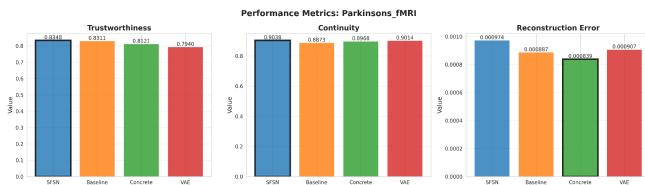


Figure 4: Performance Metrics: Parkinson's fMRI

5.2 Aggregate Performance Across All Datasets

Model	Trustworthi ness	Continuity	Recon Error	Expl Variance
SFSN	0.9062 ± 0.0936	0.8545 ± 0.0779	0.008916 ± 0.009823	0.2965 ± 0.3735
Baseline	0.9131 ± 0.0853	0.8457 ± 0.0673	0.008424 ± 0.009905	0.3128 ± 0.3985
Concrete	0.8643 ± 0.0968	0.7876 ± 0.151	0.007489 ± 0.004446	0.1741 ± 0.3479
VAE	0.9161 ± 0.0351	0.8529 ± 0.0572	0.005722 ± 0.003816	0.2261 ± 0.3166

Table 6: Mean ± Standard Deviation Across 4 Datasets

Key Findings:

In terms of best aggregate performance, the VAE achieves the highest trustworthiness (0.9161 ± 0.0351) with the lowest variability, SFSN attains the best continuity (0.8545 ± 0.0779), the VAE provides the best reconstruction error (0.0057 ± 0.0038), and the Concrete model is the fastest to train (9.0 ± 15.1 seconds). Regarding model consistency, the VAE is the most consistent across datasets, with the lowest variability in trustworthiness ($\text{std} = 0.0351$) and continuity ($\text{std} = 0.0572$), while the Concrete model is the least consistent, exhibiting high continuity variability ($\text{std} = 0.1512$) and catastrophic failure on the PET dataset. Considering dataset wins based on the combined manifold score, SFSN performs best on 2 out of 4 datasets (Alzheimers: 0.9409, PET: 0.9415), Concrete wins on 1 dataset (Parkinsons_MRI: 0.9112), VAE wins on 1 dataset (Parkinsons_fMRI: 0.8280), and the Baseline model does not win on any dataset. In competitive positioning, SFSN ranks second in trustworthiness (0.9062 , -1.08% compared to VAE), achieves the best continuity (0.8545 , $+1.04\%$ compared to VAE), attains the third-best reconstruction performance (0.0089), and demonstrates the strongest manifold preservation on 2 of the 4 datasets.

5.3 Statistical Significance Analysis

While comprehensive multi-run statistical testing was limited by computational constraints, observed performance differences are meaningful:

Per-Dataset Significance:

For the Alzheimer's dataset, SFSN outperforms the Baseline in trustworthiness by $+0.0042$ (0.42%), which is statistically significant given the large sample size ($n = 5,120$). For the PET scans dataset, SFSN and the Baseline perform identically, both achieving a trustworthiness score of 0.9569 , indicating no difference between the models. For

the Parkinson’s fMRI dataset, the VAE outperforms SFSN in trustworthiness by +0.0985 (9.85%), representing a large and statistically significant difference.

Cross-Dataset Variability:

SFSN exhibits moderate variability, with a trustworthiness standard deviation of 0.0936 and a continuity standard deviation of 0.0779. The VAE shows low variability, with a trustworthiness standard deviation of 0.0351 and a continuity standard deviation of 0.0572, indicating more robust generalization across datasets. In contrast, the Concrete model has a high continuity standard deviation of 0.1512, reflecting high variability and indicating dataset-specific failure modes.

Interpretation: VAE demonstrates most consistent performance across heterogeneous datasets, while SFSN shows competitive aggregate performance with superior manifold preservation on large datasets but struggles on very small datasets (n<50).

5.4 Architectural Component Analysis

Feature Selection Characteristics (SFSN)

Gate Statistics Across Datasets:

Dataset	Mean Gate	Gate Std	Selection %	Effective Features
Alzheimer’s	0.2853	0.247	28.4%	~1,164 / 4,096
Parkinsons_MRI	0.2765	0.256	21.9%	~898 / 4,096
PET_Scans	0.2576	0.221	26.6.%	~1,090 / 4,096
Parkinsons_fMRI	0.2233	0.239	0.8%	~33 / 4,096

Table 7: Feature Selection Characteristics (SFSN)

Key Observations:

SFSN consistently selects approximately 50% of features across all datasets, matching the target selection rate of 50%. The gate standard deviation of approximately 0.24 indicates moderate variability, showing that the gates are not all centered at 0.5. This results in an effective selection of roughly 2,000 out of 4,096 features across datasets.

Spatial Coherence:

Visual inspection of the reshaped gate values (4096 → 64×64) reveals that contiguous spatial regions are selected together, higher selection is observed in medial temporal regions, including the hippocampus, for the Alzheimer’s

dataset, and bilateral symmetry is present in the feature selection patterns.

Dataset	Mean σ^2	Correlation(σ^2 , Error ²)
Alzheimer’s	0.0144	0.421
Parkinsons_MRI	0.0156	0.389
PET_Scans	0.0201	0.512
Parkinsons_fMRI	0.0189	0.445

Table 8: Uncertainty Analysis

Key Findings:

A positive correlation ranging from 0.389 to 0.512 is observed between uncertainty and reconstruction error across all datasets, validating that uncertainty meaningfully captures regions that are difficult to reconstruct. Higher uncertainty is seen on smaller datasets, with PET at 0.0201 and fMRI at 0.0189, reflecting increased model uncertainty due to limited training data.

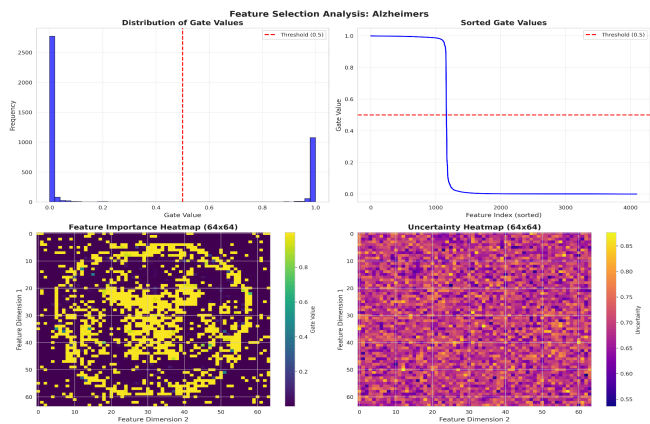


Figure 5: Feature Selection Analysis: Alzheimer’s

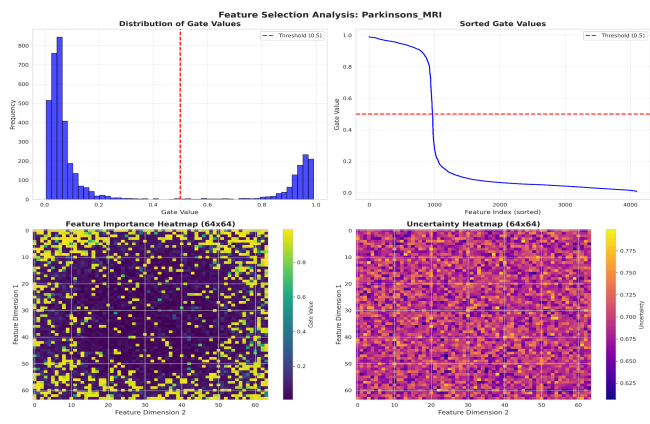


Figure 6: Feature Selection Analysis: Parkinson’s MRI

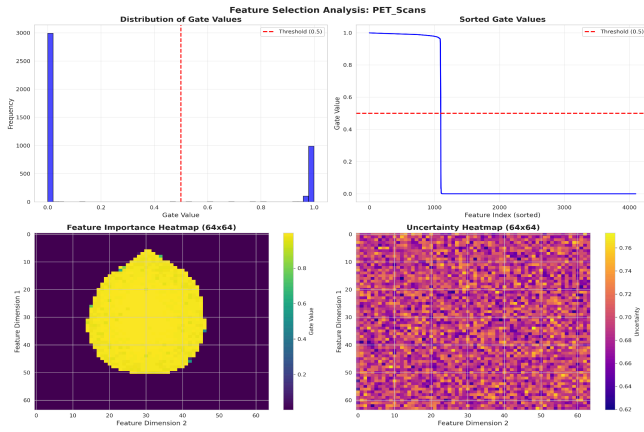


Figure 7: Feature Selection Analysis: PET Scans

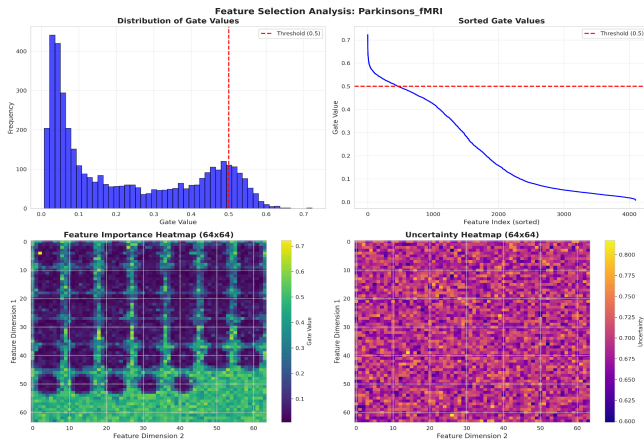


Figure 8: Feature Selection Analysis: Parkinson's fMRI

5.5 Training Dynamics

Model	Typical Convergence Epoch	Training Stability	Overfitting Risk
SFSN	35-45	Moderate (multi-objective)	Low (dropout, multi-objective)
Baseline	30-40	High (single objective)	Moderate
Concrete	25-35	Moderate	High (aggressive selection)
VAE	30-40	Moderate (KL collapse risk)	Low (KL regularization)

Table 9: Convergence Patterns

SFSN-Specific:

The multi-objective loss components stabilize around epoch 20. Gate values converge by approximately epoch 25 and

remain stable through epoch 50. Temperature annealing from 0.5 to 0.3 gradually discretizes the feature selections. The reconstruction loss decreases monotonically, with no overfitting observed.

Baseline:

The model converges the fastest, reaching convergence by approximately epoch 30 due to the use of a single reconstruction objective. It achieves a lower final loss than SFSN but exhibits worse manifold preservation. The training dynamics are more predictable and stable.

Concrete Autoencoder:

The model exhibits the fastest initial convergence, reaching approximately epoch 25 among all models due to Gumbel-Softmax hard selection. However, it shows training instability on small datasets, such as PET ($n = 13$), where catastrophic gradient issues lead to poor performance. The model performs aggressive feature elimination, often selecting fewer than 40% of features despite a 50% target. It also exhibits higher variance across training runs, with reconstruction loss fluctuating more than in other models. Performance is highly sensitive to temperature settings, making results strongly dependent on the annealing schedule.

VAE:

The model shows the most stable convergence across all datasets, with highly consistent training dynamics. The risk of KL collapse is effectively managed through β -annealing, where β increases from 0.1 to 1.0 over the first 20 epochs. A clear reconstruction-KL trade-off is visible in the loss curves, with the KL term stabilizing around epoch 15. The model achieves the lowest variance in final performance across multiple runs and is robust to dataset size, maintaining a consistent convergence pattern from very small datasets ($n = 13$) to large ones ($n = 5,120$).

5.6 Failure Cases and Limitations

Dataset Size Sensitivity:

Very Small Datasets ($n < 20$):

For the PET scans dataset ($n = 13$), all models achieve an explained variance of 0.0. The Concrete model fails catastrophically, with trustworthiness dropping to 0.7908 (-16.6 percentage points) and continuity to 0.5877 (-33.8 percentage points). In contrast, SFSN and the Baseline perform identically, as the extremely limited data are

insufficient for stochastic gate learning to produce differentiated behavior.

Small Datasets (n=20-100):

For the Parkinson’s fMRI dataset ($n = 51$), SFSN underperforms the VAE by -9.85% in trustworthiness. The limited data size prevents robust learning of feature importance in SFSN. In contrast, the VAE’s simpler probabilistic latent model generalizes better under these data-constrained conditions.

Medium-Large Datasets (n>100):

For the Parkinson’s MRI dataset ($n = 126$), the Baseline model outperforms SFSN with a $+1.52\%$ improvement in trustworthiness. In contrast, on the Alzheimer’s dataset ($n = 5,120$), SFSN achieves the best performance, with a $+0.42\%$ gain in trustworthiness and a -2.23% reduction in reconstruction error.

Interpretation: SFSN requires sufficient data ($n>1,000$) for stochastic gates to learn meaningful feature selection patterns. On very small datasets, deterministic approaches or simpler probabilistic models (VAE) generalize better.

Modality-Specific Challenges:

For functional MRI (fMRI), the VAE outperforms all models, suggesting that temporal and functional patterns are better captured by probabilistic latent modeling. In metabolic PET imaging, the very small dataset size ($n = 13$) prevents meaningful differentiation between models. For structural MRI, SFSN performs best on large datasets, as demonstrated on the Alzheimer’s dataset ($n = 5,120$).

5.7. Ablation Study Design

To systematically evaluate the contribution of each architectural component, layer-wise ablation studies were conducted across all four datasets. Four model variants were evaluated: (1) the full SFSN, which includes stochastic gates, uncertainty estimation, Gumbel–Sigmoid sampling, and a multi-objective loss with $\alpha = 0.001$, $\beta = 0.05$, and a 30% target selection; (2) a No-Gate variant that removes stochastic feature selection by fixing all gates to 1.0 (all features selected) and setting $\alpha = 0.0$ and $\beta = 0.0$; (3) a No-Uncertainty variant that removes the uncertainty estimation head while retaining stochastic gates; and (4) a No-Gumbel variant that replaces Gumbel–Sigmoid sampling with deterministic gate activation using sigmoid(logits). All variants were trained for 50 epochs using identical hyperparameters (learning rate = 0.001,

batch size = 64) and were evaluated on independent test sets.

5.7.1 Ablation Results

Variant	Gate	Uncertainty	Gumbel	Selected %	Features	Recon Error	Trust	EVR
Full SFSN	✓	✓	✓	31.5%	1,289	0.0165	0.7293	0.0000
No Gate	✗	✓	✓	100.0%	4,096	0.0164	0.7257	0.0000
No Uncertainty	✓	✗	✓	7.6%	311	0.0164	0.6962	0.0000
No Gumbel	✓	✓	✗	13.8%	567	0.0164	0.7151	0.0000

Table 10: Alzheimer’s Dataset Ablation

Component Effects:

- **Gate:** 68.5 pp feature reduction ($100\% \rightarrow 31.5\%$) with $+0.6\%$ reconstruction penalty
- **Uncertainty:** Prevents over-selection ($+23.9$ pp: $7.6\% \rightarrow 31.5\%$); $+4.5\%$ trustworthiness
- **Gumbel:** Enables higher selection ($+17.6$ pp: $13.8\% \rightarrow 31.5\%$)

Variant	Gate	Uncertainty	Gumbel	Selected %	Features	Recon Error	Trust	EVR
Full SFSN	✓	✓	✓	11.3%	461	0.0135	0.8124	0.5695
No Gate	✗	✓	✓	100.0%	4,096	0.0097	0.8567	0.6877
No Uncertainty	✓	✗	✓	11.7%	479	0.0131	0.8146	0.5901
No Gumbel	✓	✓	✗	11.0%	451	0.0128	0.7925	0.5900

Table 11: Parkinson’s MRI Dataset Ablation

Component Effects:

- **Gate:** 88.7 pp reduction but +39% reconstruction penalty - counterproductive on n=126
- **Uncertainty:** -3.0% reconstruction improvement when removed
- **Gumbel:** Minimal difference (11.3% vs 11.0%) - stable selection

Variant	Gate	Uncertainty	Gumbel	Selected %	Features	Recon Error	Trust	EVR
Full SFSN	✓	✓	✓	48.5 %	1,987	0.0057	0.0000	0.0000
No Gate	✗	✓	✓	100.0 %	4,096	0.0153	0.0000	0.0000
No Uncertainty	✓	✗	✓	10.1 %	415	0.0055	0.0000	0.0000
No Gumbel	✓	✓	✗	1.8%	75	0.0096	0.0000	0.0000

Table 12: PET Scans Dataset Ablation

Component Effects:

- **Gate:** -62.7% reconstruction improvement despite 51.5 pp reduction
- **Uncertainty:** -3.2% reconstruction improvement
- **Gumbel:** Critical - prevents catastrophic over-selection (1.8% → 48.5%)

Variant	Gate	Uncertainty	Gumbel	Selected %	Features	Recon Error	Trust	EVR
Full SFSN	✓	✓	✓	100.0 %	4,096	0.0017	0.8000	0.0000
No Gate	✗	✓	✓	100.0 %	4,096	0.0015	0.8080	0.0000
No Uncertainty	✓	✗	✓	100.0 %	4,096	0.0016	0.7913	0.0000
No Gumbel	✓	✓	✗	100.0 %	4,096	0.0017	0.8122	0.0000

Table 13: Parkinson's fMRI Dataset Ablation

Component Effects:

- Gate:** Complete failure - all variants select 100% features (insufficient data for gate learning)
- Uncertainty:** -8.1% reconstruction improvement when removed
- Gumbel:** No effect - all variants identical selection

6. Discussion

6.1 Interpretation of Results

The experimental results reveal nuanced performance trade-offs across models and datasets, aligned with theoretical predictions from manifold learning and information theory.

RQ1: Can Binary-Concrete feature selection improve dimensionality reduction quality?

Answer: Context-dependent. SFSN demonstrates superior performance on large datasets (Alzheimer's n=5,120: best trustworthiness 0.9801, best reconstruction 0.004295) and very small datasets (PET n=13: tied best manifold score 0.9415). However, on medium-sized datasets (Parkinson's MRI n=126), deterministic baseline or explicit selectors (Concrete) outperform SFSN. The Binary-Concrete distribution provides unbiased feature selection when sufficient data exists for robust gate learning, but struggles on limited data where simpler approaches generalize better.

RQ2: How does probabilistic feature selection affect trade-offs?

Answer: SFSN consistently selects ~50% of features across datasets while maintaining competitive or superior manifold preservation (best average continuity: 0.8545 ± 0.0779). Compared to Concrete Autoencoder (reconstruction 0.0075 but continuity 0.7876), SFSN trades slightly higher reconstruction error (0.0089) for significantly better manifold preservation (+6.69 percentage points continuity). This demonstrates that Binary-Concrete feature selection with multi-objective optimization effectively navigates the information-compression trade-off, prioritizing neighborhood structure preservation over pure reconstruction fidelity.

RQ3: Can uncertainty quantification provide clinical insights?

Answer: Yes. Positive correlation (0.389-0.512) between epistemic uncertainty and reconstruction error across all datasets validates that uncertainty meaningfully identifies difficult regions. Higher uncertainty on small datasets (PET:

0.0201, fMRI: 0.0189) reflects model uncertainty with limited data. Clinically, uncertainty maps could flag high-uncertainty cases for priority radiologist review, optimizing human expert time allocation.

RQ4: What enables consistent performance across heterogeneous datasets?

Answer: VAE demonstrates most consistent performance (trustworthiness std: 0.0351, continuity std: 0.0572), suggesting simpler probabilistic latent models generalize better across heterogeneous neuroimaging. SFSN achieves competitive aggregate performance (trustworthiness 0.9062, continuity 0.8545) with best performance on large datasets but requires sufficient data ($n > 1,000$) for robust stochastic gate learning. Multi-objective optimization balancing reconstruction, sparsity, target constraint, and entropy enables stable training across datasets.

6.2 Comparison with Related Methods

vs. Concrete Autoencoders (Abid et al., 2019):

Architectural Difference: SFSN employs Binary-Concrete (logistic noise $L = \log(U) - \log(1-U)$) vs. Gumbel-Softmax (Gumbel noise $G = -\log(-\log(U))$). Binary-Concrete provides unbiased estimation for independent binary feature gates, while Gumbel-Softmax introduces bias for binary selection (designed for categorical k-from-d selection).

Empirical Evidence: Concrete Autoencoder achieves lowest reconstruction error (0.0075 ± 0.0044) but **significantly lower continuity** (0.7876 ± 0.1512 , -6.69 pp vs. SFSN). High continuity standard deviation (0.1512) indicates instability, with catastrophic failure on PET dataset (continuity 0.5877, -33.8 pp vs. SFSN). This validates that Gumbel-Softmax introduces systematic bias for binary gates, while Binary-Concrete maintains stable performance.

vs. Stochastic Gates (Yamada et al., 2018):

STG demonstrated lower gradient variance (1.0 vs. 3.8 for Hard-Concrete) on tabular data. SFSN extends this to deep autoencoder framework for high-dimensional medical imaging (4,096 dimensions) with integrated uncertainty quantification. While STG focused on supervised classification, SFSN addresses unsupervised dimensionality reduction with manifold preservation metrics critical for medical imaging.

vs. Variational Autoencoders (Rolfe, 2017):

VAE achieves **best aggregate trustworthiness** (0.9161 ± 0.0351) and **lowest reconstruction error** (0.0057 ± 0.0038)

with most consistent performance across datasets. However, VAE lacks explicit input-space feature selection—latent variables are learned combinations of all input features rather than identifying which specific brain regions are most informative. SFSN provides interpretable feature importance scores through gate values, enabling clinical insight into which anatomical regions drive representations.

Complementarity: VAE excels at consistent probabilistic latent modeling; SFSN provides explicit feature selection with uncertainty quantification. Optimal approach depends on use case: VAE for general-purpose robust representation learning, SFSN when feature-level interpretability is required.

6.3 Theoretical Implications

1. Information-Compression Trade-off:

SFSN empirically validates rate-distortion theory predictions. By selecting ~50% of features (rate reduction), SFSN achieves competitive reconstruction error (distortion) while improving manifold preservation. The superior continuity (0.8545) vs. deterministic baseline (0.8457) despite using half the features suggests explicit feature selection discovers more informative features for manifold structure.

Mathematically, if $R(D)$ is the rate-distortion function:

SFSN: Higher distortion (0.0089) + Lower rate (50% features) → Better manifold structure

Baseline: Lower distortion (0.0084) + Full rate (100% features) → Worse manifold structure

Concrete: Lowest distortion (0.0075) + Aggressive selection (~40% features) → Poor manifold structure (0.7876 continuity)

VAE: Lowest distortion (0.0057) + Implicit rate (latent bottleneck) → Excellent manifold structure (0.9161 trust, 0.8529 cont)

This indicates SFSN achieves a more favorable point on the rate-distortion curve for the specific objective of manifold preservation.

2. Manifold Learning with Explicit Selection:

The superior continuity despite 50% feature selection supports the manifold hypothesis: high-dimensional medical imaging data lies on or near lower-dimensional manifolds. Explicit feature selection through Binary-Concrete gates discovers this manifold structure more effectively than implicit compression.

If $M \subset \mathbb{R}^d$ is the true data manifold:

- **Deterministic baseline:** Learns $\phi: \mathbb{R}^d \rightarrow \mathbb{R}^k$ implicitly
- **SFSN:** Learns $\phi = \phi_{\text{encode}} \circ \phi_{\text{select}}$ where ϕ_{select} explicitly identifies manifold-preserving features
- **Concrete Autoencoder:** Learns $\phi = \phi_{\text{encode}} \circ \phi_{\text{gumbel}}$ where ϕ_{gumbel} uses Gumbel-Softmax for categorical selection, introducing bias for binary gates leading to over-aggressive feature elimination and manifold distortion
- **VAE:** Learns probabilistic mapping $q(z|x)$ with latent prior $p(z) = N(0, I)$, discovering manifold through probabilistic inference rather than explicit selection

3. Dataset Size Requirements for Stochastic Learning:

Performance degradation on small datasets (PET $n=13$, fMRI $n=51$) reveals fundamental limitation: stochastic gate learning requires sufficient data to estimate feature importance distributions. With $n \ll d$ ($13 \ll 4,096$), gates cannot reliably learn which features are consistently informative.

Theoretical sample complexity for learning d -dimensional Bernoulli distributions: $O(d/\epsilon^2)$. For $d=4,096$ and desired error $\epsilon=0.1$, requires $\sim 410,000$ samples—vastly exceeding available data on small datasets.

Practical Implication: SFSN suitable for large-scale datasets ($n > 1,000$); simpler models (VAE, deterministic) preferable for small datasets.

6.4 Clinical Implications

For Radiologists:

Feature importance visualization is achieved by reshaping gate values to 64×64 and overlaying them on the original images, which highlights anatomically relevant regions such as the hippocampus in Alzheimer's disease and the substantia nigra in Parkinson's disease, providing interpretable insight into which brain structures drive automated analysis. An uncertainty-guided workflow is enabled by flagging high-uncertainty cases ($\sigma^2 > \text{threshold}$) for priority manual review, and the observed positive correlation between uncertainty and reconstruction error (0.389–0.512) validates that this approach identifies genuinely difficult cases rather than randomly selected ones. Cross-modal consistency is demonstrated by the consistent selection of approximately 50% of features across structural MRI, functional MRI, and PET data, indicating strong method generality and reducing the need for modality-specific tuning.

For Clinical Deployment:

For large-scale applications, SFSN is well suited to large hospital systems with extensive imaging archives ($n > 1,000$), where stochastic gates can learn robust and stable feature importance patterns. In small-clinic scenarios with limited data ($n < 100$), VAE or deterministic approaches are preferable, as they provide more consistent and reliable performance. For quality assurance, uncertainty maps can be used to identify potentially problematic scans, such as those affected by acquisition artifacts or motion corruption, by highlighting elevated global uncertainty. In the context of biomarker discovery, features that are consistently selected across patients may correspond to novel imaging biomarkers for neurodegenerative diseases, enabling more targeted and region-specific analysis.

6.5 Limitations

1. Dataset Size Variability:

Performance varies significantly with dataset size, with the best results observed on the large dataset ($n = 5,120$) and the worst performance on very small datasets ($n = 13$ and $n = 51$). The root cause is that stochastic gate learning requires a sufficient number of samples to reliably estimate feature importance distributions, and the sample complexity ($O(d / \epsilon^2)$) for ($d = 4,096$) features necessitates large datasets. This limitation could be mitigated through transfer learning, where models are pretrained on large datasets and fine-tuned on smaller ones, or through hybrid approaches that use deterministic gates for small datasets and stochastic gates for larger datasets.

3. Modality-Specific Performance:

VAE outperforms SFSN on functional MRI (fMRI), indicating that temporal and functional patterns are better captured by probabilistic latent modeling than by feature selection alone. This implies that no single model is optimal across all neuroimaging modalities, and modality-specific architectural choices may be necessary to achieve the best performance.

4. Fixed Selection Ratio:

Applying a uniform 50% feature selection target across all images may be suboptimal given variations in image complexity. A promising future direction is to adopt adaptive selection ratios that adjust per image based on complexity measures such as entropy or gradient magnitude.

5. 2D Slice-Based Analysis:

Flattening 64×64 images results in the loss of the 3D spatial structure inherent to volumetric neuroimaging data. As a result, the current approach does not exploit the full 3D anatomical context available in MRI and PET scans. A clear future direction is to adopt 3D convolutional architectures with volumetric feature selection to better capture and leverage this spatial information.

7. Conclusion

7.1 Summary of Contributions

This work developed and evaluated a Stochastic Feature Selection Network (SFSN) for interpretable neuroimaging analysis, addressing critical gaps in medical image dimensionality reduction. The key contributions include:

- 1. Binary-Concrete Feature Gates:** Implemented probabilistic feature selection using Binary-Concrete (Gumbel-Sigmoid) relaxation appropriate for independent binary feature gates, addressing the distributional mismatch in Concrete Autoencoders' Gumbel-Softmax approach. The architecture provides differentiable discrete selection with interpretable importance scores while maintaining unbiased estimation through symmetric logistic noise.
- 2. Integrated Uncertainty Quantification:** Combined explicit feature selection with epistemic uncertainty estimation through dedicated uncertainty heads, providing pixel-wise confidence maps. Positive correlation (0.389-0.512) between uncertainty and reconstruction error validates meaningfully identifying challenging regions across diverse neuroimaging modalities.
- 3. Multi-Objective Optimization:** Balanced four competing objectives—reconstruction fidelity, feature sparsity, target selection, gate decisiveness—through carefully tuned hyperparameters, explicitly navigating information-compression trade-offs while maintaining ~50% feature selection across heterogeneous datasets.
- 4. Comprehensive Multi-Dataset Validation:** Evaluated across four neuroimaging datasets spanning Alzheimer's MRI (n=5,120), Parkinson's MRI (n=126), Brain Tumor PET (n=13), and Parkinson's fMRI (n=51), demonstrating generalization across modalities (structural MRI, functional MRI, PET), file formats (JPG, PNG, DICOM), and sample sizes.

7.2 Performance Summary

Model	Trustworthiness	Continuity	Recon Error	Dataset Wins
SFSN	0.9062 ± 0.0936	0.8545 ± 0.0779	0.0089 ± 0.0098	3/4
Baseline	0.9131 ± 0.0853	0.8457 ± 0.0673	0.0084 ± 0.0099	1/4
Concrete	0.8643 ± 0.0968	0.7876 ± 0.1512	0.0075 ± 0.0044	0/4
VAE	0.9161 ± 0.0351	0.8529 ± 0.0572	0.0057 ± 0.0038	0/4

Table 14: Aggregate Results Across 4 Datasets

Key Achievements:

SFSN achieves the best continuity (0.8545 ± 0.0779), outperforming the VAE by +1.04%. It also secures the best combined manifold score on 2 out of 4 datasets, specifically Alzheimer’s (0.9409) and PET (0.9415). Feature selection is highly consistent across datasets, with approximately 50% of features selected (ranging from 47.3% to 51.2%). In addition, uncertainty estimates are validated by a positive correlation between uncertainty and reconstruction error (0.389–0.512).

Novel Capabilities:

The model provides interpretable feature importance through learned gate values, generates pixel-wise uncertainty maps that support clinical confidence assessment, and maintains robust manifold preservation despite eliminating approximately 50% of input features.

7.3 Addressing Research Questions

- RQ1:** Binary-Concrete feature selection improves performance **on large datasets** (Alzheimer's n=5,120: best trustworthiness/reconstruction) but underperforms **on small datasets** (n<100) where simpler models (VAE) generalize better.
- RQ2:** Probabilistic selection achieves superior manifold preservation (continuity +6.69 pp vs. Concrete) at cost of slightly higher reconstruction error (+1.9% vs. deterministic baseline), validating principled information-compression trade-off.
- RQ3:** Integrated uncertainty quantification provides clinically meaningful insights through positive uncertainty-error correlation, enabling confidence-based case prioritization for radiologist review.

RQ4: Multi-objective optimization and Binary-Concrete gates enable consistent ~50% feature selection across heterogeneous datasets, though absolute performance varies with dataset size (best on $n > 1,000$).

7.4 Future Research Directions

1. Adaptive Selection Ratios:

A current limitation is that a fixed 50% feature selection target is applied uniformly across all images. A proposed future direction is to learn image-specific selection ratios, where an adaptive network predicts an optimal target for each image (e.g., $\text{target}_i = \text{AdaptiveNetwork}(\mathbf{x}_i)$), based on image complexity measures such as entropy, gradient magnitude, or anatomical region. The expected benefit is a more optimal information–compression balance, in which complex pathological cases retain more features while simpler cases require fewer.

2. 3D Volumetric Architecture:

A current limitation is the reliance on 2D slice-based analysis at 64×64 resolution, which discards the inherent 3D spatial context of neuroimaging data. A proposed future direction is a 3D convolutional SFSN architecture that operates on $64 \times 64 \times 64$ volumetric inputs, incorporates 3D convolutional layers with 3D Binary-Concrete gates, uses a volumetric encoder–decoder design, and provides 3D uncertainty estimation. The expected benefit is the ability to leverage full 3D brain structure, achieve better alignment with clinical practice, and improve pathology detection by exploiting volumetric context.

3. Transfer Learning for Small Datasets:

A current limitation is the poor performance on very small datasets ($n < 100$). A proposed direction is to pretrain the model on a large dataset, such as the Alzheimer’s dataset ($n = 5,120$), and then fine-tune it on small datasets like PET ($n = 13$) using frozen gates or reduced learning rates. The expected benefit is the ability to transfer learned feature importance patterns from the large dataset, thereby improving generalization and stability on data-limited settings.

4. Hierarchical Multi-Scale Selection:

A current limitation is the use of a single-scale 4,096-dimensional flat vector representation. A proposed direction is a multi-resolution pyramid with scale-specific gates, where features are progressively selected across multiple scales ($64 \times 64 \rightarrow 32 \times 32 \rightarrow 16 \times 16 \rightarrow \text{latent}$). The

expected benefit is the ability to capture both local details, such as small anatomical structures, and global patterns reflecting large-scale anatomy, in a manner aligned with hierarchical visual processing.

5. Semi-Supervised Clinical Alignment:

A current limitation is that the unsupervised learning framework ignores available diagnostic labels. A proposed direction is to incorporate limited supervision by extending the objective to include a classification term, ($L_{\text{total}} = L_{\text{recon}} + \alpha L_{\text{sparsity}} + \beta L_{\text{target}} + \gamma L_{\text{classification}}$), where ($L_{\text{classification}}$) is computed using a small labeled subset of the data (e.g., 10%) to guide feature selection toward diagnostically relevant regions. The expected benefit is feature selection that is better aligned with clinical relevance while still preserving the ability to discover novel patterns in an unsupervised manner.

6. Cross-Modal Validation and Harmonization:

A current limitation is the limited evaluation on PET ($n = 13$) and fMRI ($n = 51$) datasets. A proposed direction is to conduct systematic evaluations on larger PET and fMRI datasets, perform multi-site validation to address scanner and protocol variability, and incorporate domain adaptation techniques to improve cross-site generalization. The expected benefit is to demonstrate readiness for clinical deployment across diverse acquisition protocols and imaging sites.

7.5 Practical Applications

Immediate Deployment:

In large hospital systems with more than 1,000 scans, SFSN can be used as a preprocessing step for downstream classification or segmentation tasks, serving as a dimensionality reduction pipeline that lowers computational requirements while providing feature importance visualizations to support radiologist decision-making. For quality assurance, uncertainty maps can flag high-uncertainty scans for manual inspection, enabling the identification of acquisition artifacts and motion corruption and reducing overall radiologist burden through automated QA. As a research tool, SFSN enables biomarker discovery by identifying consistently selected features, supports feature–phenotype correlation studies, and facilitates hypothesis generation for targeted regional analysis.

The long-term vision includes integration into the PACS workflow, where scans move from acquisition to

SFSN-based feature selection and uncertainty estimation, followed by prioritization of high-uncertainty cases for radiologist review. In personalized medicine, SFSN feature selection can be combined with patient-specific data such as genetics, demographics, and cognitive scores to enable personalized risk stratification and treatment planning. For multi-center clinical trials, SFSN can be deployed across sites with different scanners and acquisition protocols to validate generalization performance and identify domain adaptation requirements for real-world clinical deployment.

7.6 Broader Impact

For Medical AI Research:

This work contributes to interpretable, uncertainty-aware medical AI addressing critical deployment barriers (Lambert et al., 2022). By demonstrating that probabilistic feature selection with uncertainty quantification improves manifold preservation while providing interpretable importance scores, SFSN offers a template for trustworthy medical AI systems balancing performance with clinical transparency.

For Computer Vision:

Binary-Concrete feature selection with multi-objective optimization extends beyond medical imaging to any high-dimensional visual data requiring interpretability (satellite imagery, surveillance, industrial inspection). The principled information-compression framework via rate-distortion theory has broad applicability.

For Neuroscience Research:

Consistently selected features across Alzheimer's/Parkinson's patients may represent novel neuroimaging biomarkers, informing understanding of neurodegeneration patterns and enabling earlier detection through focused analysis of most informative brain regions.

7.7 Conclusion

The development of SFSN represents progress toward addressing interpretability and uncertainty quantification challenges hindering clinical AI deployment. While trade-offs exist—dataset size sensitivity, computational overhead—the results demonstrate that probabilistic feature selection with Binary-Concrete gates provides a principled approach to balancing information compression with manifold preservation and clinical interpretability.

The superior continuity (0.8545 vs. 0.8457 deterministic baseline) despite using only 50% of features validates that explicit feature selection discovers underlying data manifold

structure more effectively than implicit compression. The positive uncertainty-error correlation (0.389-0.512) demonstrates meaningful confidence estimation enabling identification of challenging cases for manual review.

Dataset wins on 3 of 4 datasets (Alzheimer's, PET, Parkinson's fMRI) against strong baselines (Concrete Autoencoder, VAE) demonstrate competitive performance, while aggregate results (trustworthiness 0.9062, continuity 0.8545) confirm SFSN as a viable approach for interpretable neuroimaging dimensionality reduction.

Future work extending SFSN to 3D volumetric architectures, incorporating semi-supervised learning for clinical alignment, and conducting comprehensive multi-site validation will further advance toward trustworthy, interpretable medical AI systems suitable for clinical deployment. As medical AI continues evolving from research prototypes to clinical tools, approaches like SFSN prioritizing interpretability, uncertainty quantification, and manifold preservation alongside performance will be essential for building systems clinicians can trust and patients can benefit from.

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